

TED UNIVERSITY

Faculty of Engineering

CMPE491 – Senior Project



Hybrid Image Enhancement System

Supervisor – *Fırat Akba*

Prepared by *Ahmet Emir Ceylan, Batuhan Taşdemir, Furkan Cabbar*

PROBLEM DEFINITION

- Ineffective Zoom:** Traditional methods just enlarge pixels, causing blur instead of recovering actual detail.
- AI Hallucinations:** Pure AI models often "hallucinate" fake textures, creating reliability risks for legal use.
- Lack of Trust:** "Black box" algorithms hide their logic, making it impossible to verify if the image is authentic.
- Core Objective:** We propose a transparent system that enhances clarity while maintaining strict fidelity to the original data

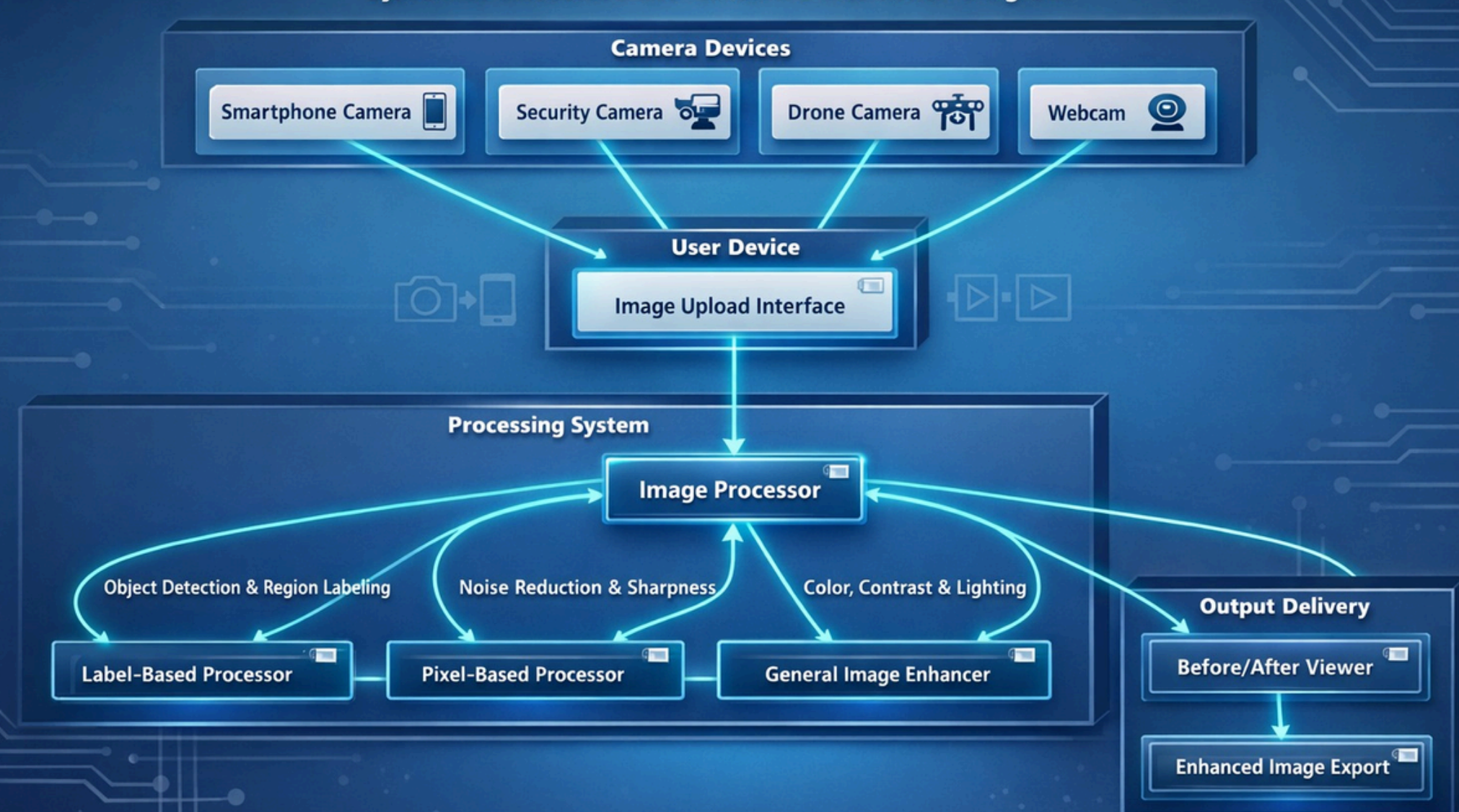
OBJECTIVES

- Hybrid Development:** We are building this system using Python and TensorFlow.
- Layered Architecture:** We utilize a modular design that integrates Pixel, Label-Based, and Global processing layers.
- Realistic Quality:** Our goal is to improve zoom clarity while keeping the image physically plausible.
- Controlled AI:** We apply strict constraints to ensure transparency and device independence.

SYSTEM ARCHITECTURE

Built on a Python framework, our pipeline integrates Semantic, Pixel, and Global processing layers. A central orchestrator restricts TensorFlow-based AI to specific analytical tasks—like edge and object detection—rather than full image synthesis. This 'controlled AI' approach prevents hallucinations, ensuring the final output is deterministic, verifiable, and strictly faithful to the original input.

System Architecture and Hardware Interaction Diagram



METHODOLOGY

Top-Down Strategy: We adopted a modular, top-down development approach, ensuring that the architectural vision guides component implementation.

Isolated Integration: Each subsystem is designed and tested independently before integration, allowing us to isolate errors and ensure stability.

ODBC Logging: We utilize optional ODBC-based logging to record system metadata, ensuring the results are reproducible and available for detailed performance analysis.

ALGORITHMS

Hybrid Integration: Integrates TensorFlow-based semantic models with deterministic image processing logic.

Guided Enhancement: Utilizes semantic labels to direct region-aware adjustments, preventing uniform over-processing.

Visual Stability: Applies pixel-level cleaning and global balancing to ensure the final output remains natural and artifact-free.



EXPECTED OUTPUTS

The system delivers high-fidelity enhancements that restore sharpness while strictly avoiding artificial 'hallucinations.' By enforcing deterministic processing, we ensure every detail remains physically plausible and verifiable. This guarantees consistent, reliable performance across all camera hardware, making the output trustworthy for safety-critical applications.

CONCLUSION

We developed a Python-based hybrid framework that merges deterministic image processing with targeted TensorFlow analysis. Unlike typical generative models, our system enforces strict architectural controls to prevent hallucinations and ensure explainability. The design prioritizes platform independence and scalability—supported by optional ODBC data management—demonstrating that trustworthy image enhancement is best achieved through disciplined, transparent engineering.

HYBRID IMAGE ENHANCEMENT

DEVICE-INDEPENDENT, EXPLAINABLE, & RELIABLE

